

The Oregon Health Insurance Experiment (OHIE) provided the first experimental evidence on the effects of Medicaid coverage on beneficiaries. The researchers did not observe significant effects of Medicaid coverage on any of the collected health measures (blood pressure, cholesterol, and blood sugar levels) (Baicker et al., 2013). Using administrative data, they also found no evidence that Medicaid coverage led to a reduction in mortality during the 16 months following coverage gain. Their estimate suggested a 16 percent reduction in mortality associated with acquiring Medicaid, but with a large confidence interval that could not rule out sizeable changes in either direction (Finkelstein et al., 2012).

As the data become available, researchers are beginning to evaluate the mortality effects of the ACA Medicaid expansions.³ A small number of recent studies use population-level mortality data to estimate changes in adult mortality in expansion states compared to non-expansion states. These studies reach very different conclusions, either unable to detect mortality effects (Black et al., 2019) or estimating varying sized reductions in adult mortality: 3.6 percent among adults ages 20-64 (Borgschulte and Vogler, 2020) and 1.2 percent among adults ages 55-64 (Chen, 2019). Yan et al. (2020) find no reduction in all-cause mortality among adults ages 20-64, but a 2.7 percent decrease in health care amenable mortality. Khatana et al. (2019) focus on changes in cardiovascular disease-related mortality among adults ages 45-64 and document a 2.9 percent reduction over the baseline mortality rate. Finally, Swaminathan et al. (2018) find an 8.5 percent decrease in one-year mortality for patients with end stage renal disease initiating dialysis associated with the ACA Medicaid expansions.

There is also recent evidence that health insurance coverage (inclusive of Medicaid) can affect mortality. Goldin et al. (2021) provide new experimental evidence documenting such mortality effects by studying an IRS program that sent informational letters about health insurance enrollment to a randomly-selected sample of taxpayers who were subject to the ACA mandate penalty for being uninsured. The letters both significantly increased enrollment in insurance (both private exchange coverage and Medicaid) and reduced mortality among those between the ages of 45 and 64 and uninsured at the time of the intervention. The results from this experiment suggest that one additional month of insurance coverage lowered the two year mortality rate of recipients by about 0.18 percentage points, or over 10 percent. The authors found no effects of the letter on mortality for those under age 45.

One of the primary impediments to fully understanding the impact of Medicaid on mortality has been data availability. Previous studies using aggregated data rely on changes in survival for the Medicaid eligible to translate into overall mortality effects observable at the population level. Data from death certificate records contain very little socioeconomic information on the decedent; in particular, they contain no information on the decedent's income, use of public programs, whether he or she previously had health insurance coverage, or other characteristics that might affect Medicaid eligibility. Without data that links information on individual Medicaid eligibility to mortality, researchers must rely on differences in exposure over larger population groups – for example, residents of certain states or counties – which contain many individuals who are not directly affected by Medicaid policy.⁴ This

³A separate but related literature has examined the relationship between public health insurance and child mortality using variation in exposure tied to the introduction of Medicaid and later expansions in public coverage under Medicaid and the Children's Health Insurance Program. For the most part, these studies have found significant declines in mortality associated with expanded coverage for infants and children both in the short-term (e.g. Currie and Gruber, 1996a,b; Goodman-Bacon, 2018; Howell et al., 2010) and long-term (Brown et al., 2018; Goodman-Bacon, 2016; Wherry and Meyer, 2016).

⁴Variation in exposure to the policy based on state of residence, as well as local geographic characteristics (such as

decreases the power to detect changes in mortality of a plausible magnitude, leading some researchers to conclude that “it will be extremely challenging for a study [on the ACA Medicaid expansions] to reliably detect effects of insurance coverage on mortality unless these data can be linked at the individual level to large-sample panel data” (Black et al., 2019).

In this paper, we aim to contribute to this knowledge base by providing new evidence on the mortality effects of Medicaid. We build on the accumulation of evidence from prior studies but offer three important advantages. First, we use individual characteristics to study mortality among the Medicaid-eligible population, rather than use analyses that rely on population-level data and may be underpowered to detect effects. This may be particularly salient in the case of the ACA Medicaid expansions, when insurance coverage is estimated to have increased by as little as 1 percentage point among all nonelderly adults (Black et al., 2019). Second, using administrative Medicaid enrollment information, we confirm a large change in coverage for our study population. The longitudinal enrollment data improves on the more commonly used survey measures of Medicaid coverage, which can be subject to substantial misreporting (Boudreaux et al., 2013), and also allow us to examine the total amount of coverage accumulated by residents of expansion states over the entire sample period. Documenting changes in accumulated coverage is important because studies suggest that health insurance coverage can have longer term effects that extend beyond just the coverage period (e.g. Boudreaux et al., 2016; Brown et al., 2018; Currie et al., 2008; Goodman-Bacon, 2016; Thompson, 2017; Wherry and Meyer, 2016; Wherry et al., 2017). Third, we estimate effects for this Medicaid-eligible population using much larger sample sizes than were used in the only experimental evaluation of Medicaid coverage, the Oregon Health Insurance Experiment. This larger sample is particularly important when examining mortality as an outcome since it is an extreme and infrequent health event. These advantages overcome the existing limitations in both experimental and non-experimental studies of Medicaid, allowing us to further advance the evidence on the program’s mortality effects.

III Data and Outcomes

To conduct our analysis, we link individuals across three different data sources. First, we use restricted data from the 2008 to 2013 waves of the American Community Survey, which has a sample size approximately 50 percent larger than the public-use version, to identify our population of interest. We select respondents who, based on their pre-ACA characteristics, were likely to benefit from the ACA Medicaid expansions. We include only individuals who either are in families with income at or under 138 percent of the FPL or who have less than a high school degree.⁵ Since we only have information on income captured at one point in time, the latter criterion is used to identify individuals who are of low socioeconomic status but might not meet the income cutoff at the time of the ACS interview. Results are similar if we include only those with less than a high school degree, or only those with incomes under 138 percent of the FPL, rather than defining the sample using the union of these two criteria; see

baseline uninsurance or poverty rates at the county level) are sources that have been used. In addition, some researchers have used information on the decedent’s educational attainment as a proxy for individual income, although this information is reported by the next of kin and subject to substantial measurement error (see e.g., Rostron et al., 2010; Sorlie and Johnson, 1996).

⁵We define family income using the Census family definition of all related individuals living in the same household. Results are very similar if we follow the more restrictive definition of a family unit proposed by the State Health Access Data Assistance Center (2012) for studies of health insurance coverage.

later discussion in Section VII. We exclude non-citizens, many of whom are not eligible for Medicaid, and those receiving Supplemental Security Income (SSI), who are automatically eligible for Medicaid in most states. We restrict our primary analysis to individuals who were ages 55 to 64 in 2014. This higher age group has relatively high mortality rates, and is also consistent with the sample criteria used in Black et al. (2019). We present results for all non-elderly adults, and for a variety of different age subgroups, in additional analyses in Section VII. Finally, we exclude residents of 4 states and DC that expanded Medicaid to low-income adults prior to 2014.⁶ There are approximately 566,000 respondents who meet our sample criteria.⁷

Descriptive statistics for the sample by state Medicaid expansion status are reported in Appendix Table A1 (all appendix information is included in an Online Appendix). The average age of the respondents in the two groups is similar. However, individuals in expansion states are slightly better off with higher average income (147% of the FPL vs. 140%) and educational attainment (45.3% with less than high school education vs. 46.8%), as well as lower baseline rates of uninsurance (32.6% vs. 37.3%), than individuals in non-expansion states. In addition, individuals in expansion states are more likely to be white or Hispanic, while a higher share of those in non-expansion states are Black.

Second, we link these data to the Census Numident file. The Census Numident file is derived from the Social Security Administration (SSA) Numerical Identification file, which includes information on date and county of birth and date of death (if it has occurred) for individuals with a Social Security Number (SSN).⁸ These data have been used in, e.g., Brown et al. (2018); Chetty et al. (2011, 2016); Dobbie and Song (2015); Sullivan and von Wachter (2009), and other research relying on death information from tax records. Total deaths reported in the SSA file by age and year closely track the numbers reported by the National Center for Health Statistics (Chetty et al., 2016). The Census Bureau updates its Numident file each year with new information from the SSA Numerical Identification file. It formats the data so that there is a single record per individual, reflecting the most accurate and up-to-date information at that point in time. We use data from the Census Numident for deaths occurring in 2017 and earlier.

The Census Numident and ACS data are linked via the Census Bureau’s Personal Identification Validation System (PVS). This system assigns individuals in each dataset a protected identification key (PIK), an anonymized identifier that allows Census to track individuals across datasets. Approximately 90 percent of all ACS respondents are successfully assigned a PIK using available information on name, address, and date of birth, with a slightly higher match rates for citizens (92 percent) (Bond et al., 2014). The assignment of a PIK allows respondents in the ACS to be matched to the Census Numident file. PIKS for the Census Numident file are assigned using social security numbers (SSNs) and date of birth (Mulrow et al., 2011). Since our analysis is restricted to older citizens, and since nearly all American citizens have SSNs assigned by the time they reach adulthood (see Bernstein et al., 2018),

⁶DE, MA, NY, and VT all expanded coverage to individuals with incomes reaching the poverty line or greater prior to the ACA; DC received approval to implement its ACA Medicaid expansion early with enrollment starting in 2011.

⁷Note that Census disclosure rules prohibit the disclosure of exact sample sizes and require rounding. All sample sizes for these data are therefore rounded according to disclosure rules.

⁸In addition to this death information from the SSA, the Census Bureau also has information on date of death from the National Death Index (NDI) for some individuals and years, which it incorporates into its date of death measure when available. The NDI collects detailed information on deaths from state vital statistics offices. Respondents to the 2008 ACS were linked to the NDI for the years 2008-2015, as part of the Mortality Disparities in American Communities project.

we expect to have nearly full coverage of deaths in the Numident file.

Once these data are linked, we observe the vital status of each individual during the year they respond to the ACS and each subsequent year. For example, we observe the vital status of an individual who responds to the 2008 ACS during each year from 2008 through 2017; for an individual who responds to the 2013 ACS, we observe his or her vital status from 2013 through 2017. We construct our outcome measure to represent mortality during each calendar year. If the individual is alive in a given year, the outcome variable takes a value of 0; if that individual died in that year it takes a value of 1. Once an individual has died, he or she is removed from the sample for subsequent years. In this way, we measure changes in the annual probability of death during a given year among individuals who were alive at the beginning of that year. The annual mortality rate is about 1.4 percent for our sample on average across all years, and approximately 1.3 percent among respondents in expansion states during the year just prior to expansion.⁹

Third, we further link our sample of ACS respondents to administrative records on Medicaid enrollment from the Centers for Medicare & Medicaid Services (CMS). We use the same PVS methods as were applied to the mortality data to link these newly available records to our sample of ACS respondents. About 93 percent of Medicaid enrollment records successfully receive a PIK assignment using SSN and date of birth information (Fernandez et al., 2015). CMS data on enrollment allows us to observe Medicaid enrollment longitudinally for our sample and to document how the probability of enrollment, and total cumulative exposure to Medicaid, changed in the expansion states relative to the non-expansion states. We use data from 2008 through 2016, the most recent data available. In our analyses of cumulative enrollment, we impute the number of days enrolled in 2017 using enrollment information from 2016, state of residence, age, gender, and race. To construct this imputation, we use information on transitions into and out of Medicaid observed in the previous year. See Appendix Section 1 for more information.¹⁰ When examining the probability of being enrolled any time during a calendar year, we simply omit 2017 and use data on enrollment through 2016.

While our data uniquely offer the opportunity to link mortality, Medicaid enrollment, and economic variables at the individual level, there are also several important limitations. First, we observe the economic characteristics of individuals (income and educational attainment, receipt of social services, and citizenship status) at the time they respond to the ACS in the pre-period, between 2008 and 2013. These are time-varying characteristics, however, and may not accurately reflect economic characteristics at the time of the Medicaid expansions for some members of our sample. For example, an individual in a low-income family in 2008 may be in a higher-income family by 2014, at the time the expansions occurred. Similarly, individuals may migrate to different states between the time they responded to the ACS and the time the expansions occurred, resulting in our misclassification of whether that individual was exposed to the eligibility expansion.¹¹ In general, we expect that this type of misclassification will bias our estimates towards zero. In addition, the CMS administrative enrollment information allows us to accurately examine and characterize the amount of exposure to the policy for our analytic sample

⁹These annual averages are calculated excluding mortality rates for individuals during their year of ACS interview.

¹⁰There are a number of other minor data issues surrounding the CMS enrollment data, which are outlined in this Appendix section.

¹¹Note, however, that individual migration decisions do not appear to be correlated with state Medicaid expansion (Goodman, 2017).